

Structured practice paper 2

Higher challenge and mixed reasoning - 50 marks - 75 minutes

YOUR AIM TODAY

Complete a realistic structured paper on quadratics, functions, coordinate geometry and circles.

YOUR RULE

Closed book. Calculator allowed. Answer all eight questions. Show essential working and stop after 75 minutes. Keep the solution section closed.

Student: Rohan

Start time: _____ Finish time: _____

Date: _____

INSTRUCTIONS

Candidate information

This paper uses linked parts and explicit mark allocations like an examination.

Time	75 minutes
Marks	50
Questions	8 compulsory structured questions
Method	Show algebra and exact values; unsupported answers may lose marks
Checking	Reserve the final 7 minutes

Question	Topic	Marks
1	Completed square and exact roots	7
2	Parameter and number of roots	6
3	Restricted inverse	6
4	Composite equation	6
5	Line and perpendicular bisector	7
6	Circle intersections	6
7	Tangent and normal	6
8	Mixed line-circle problem	6
	Total	50

EXAM RULE

If stuck, write the relevant formula or structure, leave space and continue. Return before time ends.

Structured questions

Running total after this page: 13 marks.

QUESTION 1

Completed square and exact roots

[7 marks]

(a) Express $3x^2-12x+7$ in completed-square form. [3]

(b) Hence solve $3x^2-12x+7=0$ exactly. [2]

(c) Solve $3x^2-12x+7<0$. [2]

QUESTION 2

Parameter and number of roots

[6 marks]

The equation $x^2+(k-2)x+k=0$ has a repeated root. Find the possible values of k . [4]

(b) Hence state the values of k for which the equation has no real roots. [2]

Structured questions

Running total after this page: 25 marks.

QUESTION 3

Restricted inverse

[6 marks]

GIVEN

$h(x) = (x-1)^2 - 4$, with domain $x \geq 1$.

(a) State the range of h . [1]

(b) Find $h^{-1}(x)$. [3]

(c) Solve $h^{-1}(x) = 5$. [2]

QUESTION 4

Composite equation

[6 marks]

GIVEN

$f(x) = 3x - 1$ and $g(x) = 2 - x$.

(a) Find $fg(x)$. [2]

(b) Find $gf(x)$. [2]

(c) Solve $fg(x) = gf(x)$, explaining your result. [2]

Structured questions

Running total after this page: 38 marks.

QUESTION 5

Line and perpendicular bisector

[7 marks]

GIVEN

$A=(-3,4)$ and $B=(5,-2)$.

(a) Find the gradient of AB. [2]

(b) Find an equation of AB in integer form. [2]

(c) Find the perpendicular bisector of AB. [3]

QUESTION 6

Circle intersections

[6 marks]

GIVEN

$C: x^2+y^2+8x-2y-8=0$.

(a) Write C in completed-square form and state its centre and radius. [3]

(b) Find the exact points where C meets the x-axis. [3]

Structured questions

Running total after this page: 50 marks.

QUESTION 7

Tangent and normal

[6 marks]

GIVEN

Use circle C from Question 6 and $P=(0,4)$.

(a) Show P lies on C and find the tangent at P. [3]

(b) Find where the tangent meets the x-axis. [1]

(c) Find the normal at P. [2]

QUESTION 8

Mixed line-circle problem

[6 marks]

GIVEN

The line $y=x+1$ meets $(x-1)^2+(y-2)^2=10$.

(a) Find the exact coordinates of both intersection points. [4]

(b) Find the midpoint of the two intersection points and interpret it. [2]

STOP

Solution section locked

Do not continue until the full paper has an attempt and 75 minutes have ended.

BEFORE OPENING

Write your predicted score and identify the two questions you found hardest.

Predicted score: ____ / 50. Hardest questions:

MARKING METHOD

Use another colour. Circle the first wrong line. After reading the explanation, close the booklet and redo the entire part.

Easy worked explanations

Equivalent correct methods are acceptable.

Question 1 - Completed square and exact roots [7 marks]

Part	Easy working and answer
a	Factor 3 from the x terms: $3(x^2-4x)+7=3[(x-2)^2-4]+7=3(x-2)^2-5$.
b	Set $3(x-2)^2-5=0$. Then $(x-2)^2=5/3$, so $x=2\pm\sqrt{15}/3$.
c	The upward parabola is negative between its roots: $2-\sqrt{15}/3 < x < 2+\sqrt{15}/3$.

Question 2 - Parameter and number of roots [6 marks]

Part	Easy working and answer
a	Repeated means $D=0$: $(k-2)^2-4k=0$. Expand to $k^2-8k+4=0$. Formula gives $k=[8\pm\sqrt{64-16}]/2=4\pm2\sqrt{3}$.
b	$D=k^2-8k+4$ is negative between its roots. Therefore $4-2\sqrt{3}<k<4+2\sqrt{3}$.

Easy worked explanations

Equivalent correct methods are acceptable.

Question 3 - Restricted inverse [6 marks]

Part	Easy working and answer
a	The square is at least zero, so $h(x)$ is at least -4 . Range: $y \geq -4$.
b	Write $y = (x-1)^2 - 4$. Then $y+4 = (x-1)^2$. Since $x \geq 1$, take the positive root: $x = 1 + \sqrt{y+4}$. Thus $h^{-1}(x) = 1 + \sqrt{x+4}$.
c	$1 + \sqrt{x+4} = 5$, so $\sqrt{x+4} = 4$, $x+4 = 16$ and $x = 12$.

Question 4 - Composite equation [6 marks]

Part	Easy working and answer
a	$fg(x) = f(2-x) = 3(2-x) - 1 = 5 - 3x$.
b	$gf(x) = g(3x-1) = 2 - (3x-1) = 3 - 3x$.
c	$5 - 3x = 3 - 3x$ simplifies to $5 = 3$, which is impossible. Therefore there is no solution; the expressions differ by 2 for every x .

Easy worked explanations

Equivalent correct methods are acceptable.

Question 5 - Line and perpendicular bisector [7 marks]

Part	Easy working and answer
a	$m = \frac{-2-4}{5-(-3)} = \frac{-6}{8} = -\frac{3}{4}$.
b	Using A: $y-4 = (-\frac{3}{4})(x+3)$. Multiply by 4 and rearrange: $3x+4y=7$.
c	Midpoint = (1,1). Perpendicular gradient = $\frac{4}{3}$. So $y-1 = (\frac{4}{3})(x-1)$, or $4x-3y=1$.

Question 6 - Circle intersections [6 marks]

Part	Easy working and answer
a	$x^2+8x=(x+4)^2-16$ and $y^2-2y=(y-1)^2-1$. Hence $(x+4)^2+(y-1)^2=25$. Centre (-4,1), radius 5.
b	On the x-axis $y=0$: $(x+4)^2+1=25$, so $(x+4)^2=24$. Thus $x=-4\pm 2\sqrt{6}$, giving $(-4+2\sqrt{6},0)$ and $(-4-2\sqrt{6},0)$.

Easy worked explanations

Equivalent correct methods are acceptable.

Question 7 - Tangent and normal [6 marks]

Part	Easy working and answer
a	From centre $(-4,1)$ to P, the change is $(4,3)$, length 5, so P lies on C. Radius $m=3/4$, so tangent $m=-4/3$: $y-4=(-4/3)x$, or $4x+3y=12$.
b	Set $y=0$ in $4x+3y=12$: $x=3$, so the point is $(3,0)$.
c	The normal follows the radius, $m=3/4$, through P: $y-4=(3/4)x$.

Question 8 - Mixed line-circle problem [6 marks]

Part	Easy working and answer
a	Substitute $y=x+1$, so $y-2=x-1$. Then $2(x-1)^2=10$, hence $(x-1)^2=5$ and $x=1\pm\sqrt{5}$. Since $y=x+1$, the points are $(1+\sqrt{5}, 2+\sqrt{5})$ and $(1-\sqrt{5}, 2-\sqrt{5})$.
b	Average coordinates: $(1,2)$. This is also the circle's centre, so the line passes through a diameter.

Turn marks into the next lesson

Do not record only the final answer.

Q/part	First wrong line	Error type	Correct rule

Topic	Score	Available	%
Quadratics		13	
Functions		12	
Coordinate geometry		7	
Circles		18	
Total		50	

RETEST

Redo every lost mark tomorrow without looking. Any topic below 60% returns to the notes booklet first.